Chapter 10

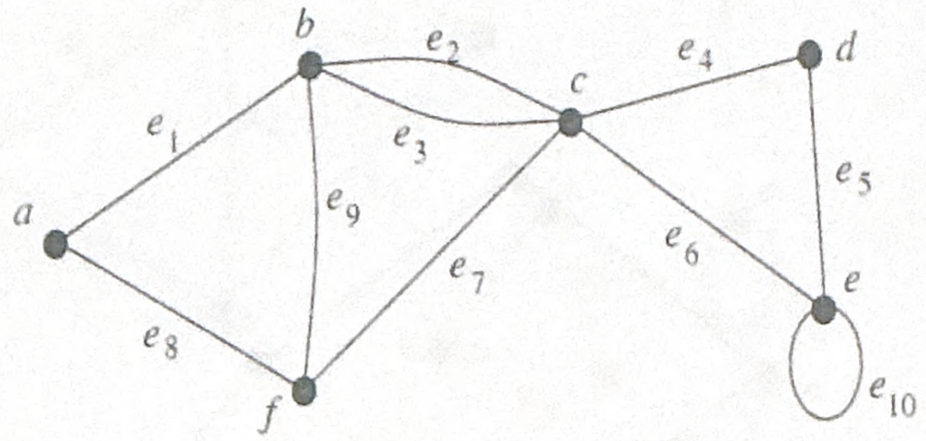
Connected Graphs

10.1 Walks and Circuits

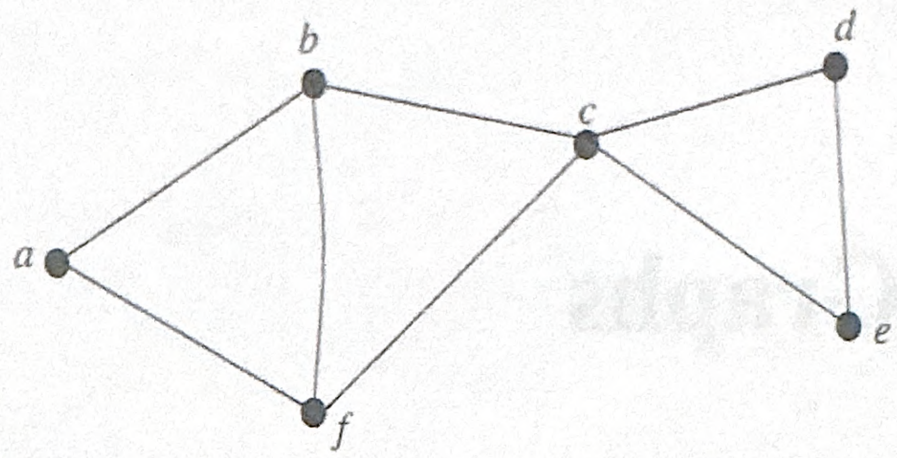
Definition: Let $G=(V,E,\psi)$ be an undirected graph. Let $u, v \in V$. An ***u-v walk*** in G is an alternating sequence of vertices and edges of G, $w_0, e_1, w_1, \cdots, w_{k-1}, e_k, w_k$, such that $w_0 = u$ and $w_k = v$ and $\psi(e_i) = \{w_{i-1}, w_i\}, 1 \leq i \leq k$. The vertices $w_i, 1 \leq i \leq k-1$ are called the ***internal vertices*** while the vertices $w_0 = u$ and $w_k = v$ are called the ***terminating vertices*** of the walk. The u-v walk is said to be connecting vertices u and v. An u-v walk is a ***closed walk*** if $u = v$. It is a ***null walk*** if k=0.

Remark: A null walk is a walk without any edge. We customarily omit the parenthesis and the commas in the representation of a walk and simply write it as $w_0e_1w_1 \cdots w_{k-1}e_kw_k$. The notation $P : w_0e_1w_1 \cdots w_{k-1}e_kw_k$ denotes $w_0e_1w_1 \cdots w_{k-1}e_kw_k$ is an $w_0 - w_k$ walk represented by P. If G is a simple graph, then as between every two adjacent vertices, there is exactly one edge joining them, the e_i's in the walk can be omitted.

Figure 10.1.6



$c\ e_2\ b\ e_9\ f\ e_7\ c\ e_4\ d\ e_5\ e\ e_{10}\ e\ e_5\ d$ is a walk in the above graph.



cbfcded is walk in the above simple graph.

Definition: A ***u-v trail*** is a *u-v* walk which does not repeat any edge.

A ***u-v path*** is a *u-v* trail which does not repeat any vertex.

A ***circuit*** is a non-null *u-v* trail in which *u=v*.

A ***cycle*** is a circuit which does not repeat any vertex except the first and the last.

Formally, let $w_0, e_1, w_1, \cdots, w_{k-1}, e_k, w_k$ be a *u-v* walk (i.e. $w_0 = u, w_k = v$).

- If $e_i \neq e_j, \forall i, j, 1 \leq i < j \leq k$, then the *u-v* walk is a *u-v trail.*
- If $w_i \neq w_j, \forall i, j, 0 \leq i < j \leq k$,. then the *u-v* walk (which is obviously a *u-v* trail) is a *u-v path.*
- If $e_i \neq e_j, \forall i, j, 1 \leq i < j \leq k$ and $w_0 = w_k$, where $k > 0$, then the *u-v* walk is a *circuit.*
- If $w_i \neq w_j, \forall i, j, 0 \leq i < j \leq k-1$, and $w_0 = w_k$. then the *u-v* walk is a *cycle.*

Example 10.1.4 In the first graph in Figure 10.1.6,

$c\ e_3\ b\ e_9\ f\ e_7\ c\ e_4\ d\ e_5\ e\ e_6\ c$ is a *c-c* trail;

$d\ e_4\ c\ e_2\ b\ e_9\ f\ e_8\ a$ is a *d-a* path;

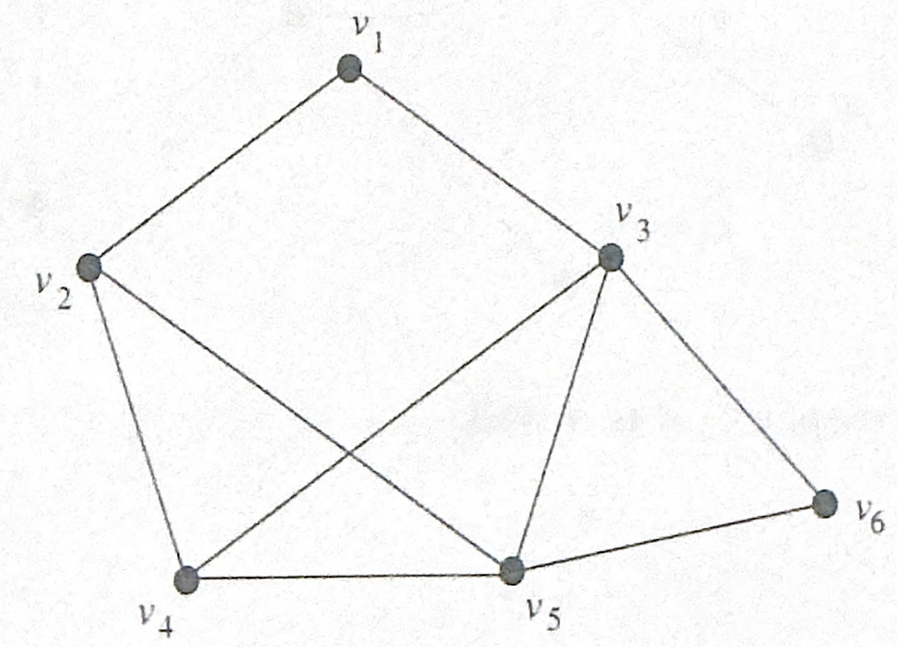
a is a null path

$c\ e_2\ b\ e_9\ f\ e_7\ c\ e_4\ d\ e_5\ e\ e_6\ c$ is a circuit;

$c\ e_3\ b\ e_9\ f\ e_7\ c$ and $e\ e_5\ d\ e_4\ c\ e_6\ e$ are cycles.

Remark: In a simple graph, every cycle contains *at least* three edges. □

Example 10.1.5



In the above simple graph,

$v_2\ v_5\ v_3\ v_6\ v_5\ v_4\ v_2$ is a circuit which is not a cycle;

$v_6\ v_5\ v_4\ v_3\ v_5$ is a trail which is not a path;
$v_1\ v_2\ v_4\ v_5$ is a path;
$v_2\ v_5\ v_3\ v_4\ v_2$ is a cycle;
$v_6\ v_3\ v_1\ v_2\ v_5\ v_3\ v_6$ is neither a circuit nor a cycle as it repeats the edge (v_3, v_6). □

Lemma 10.1.1: *Every u-v walk contains a u-v path.*

Proof: The shortest *u-v* walk within the given *u-v* walk is a *u-v* path. ■

Lemma 10.1.2: *Every circuit contains a cycle.*

Proof: The shortest circuit in a circuit is a cycle. ■

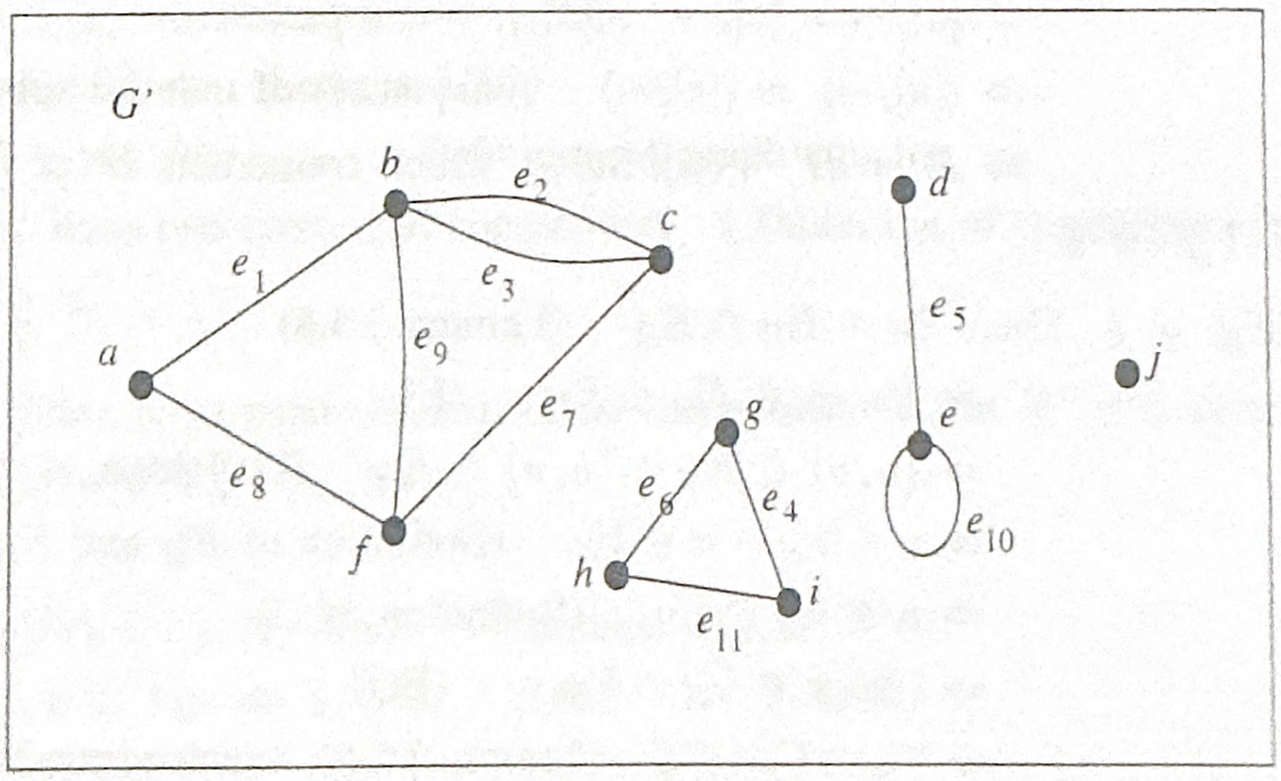
10.2 Connected Graphs

Definition: Two vertices u and v in a graph G are ***connected***, denoted by $u \sim_G v$, if there exists a *u-v* path in G. A graph is a ***connected graph*** if every two vertices in it are connected; i.e. $(\forall u \in V)(\forall v \in V)u \sim_G v$ (the subscript G can be omitted if no confusion could occur). A ***disconnected graph*** is a graph that is not connected; i.e. $(\exists u \in V)(\exists v \in V)u \not\sim_G v$.

The notation $u \not\sim_G v$ denotes $\sim \left(u \sim_G v\right)$. i.e. vertices u and v are not connected in G.

Example 10.2.6 The graph in Example 10.1.6 is a connected graph. The graph G' in Figure 7 is a disconnected graph.

Figure 7 A disconnected graph



Since adding or removing self-loops or parallel edges from a graph does not affect the connectivity of the graph, *we shall assume without loss of generality that all the graphs in this chapter are simple graphs.*

Theorem 10.2.3: *The binary relation $\sim$ is an equivalence relation in V_G.*

Proof: (left as an exercise) ■

Definition: A ***connected component*** of G is a subgraph of G induced by an equivalence class w.r.t. $\sim$. The number of components of G is denoted by $\omega(G)$ (G is omitted if no confusion could occur).

Example 10.2.7 For the graph G' in Example 7, $\omega(G') = 4$.

Lemma 10.2.4: *$\omega(G) \geq 1$, for every graph $G = (V, E)$.*

Proof: By definition, $V \neq \emptyset \Rightarrow \exists v \in V$ (Lemma 3.5.8)

$\Rightarrow (v, v) \in \sim$ ($\sim$ is reflexive)

$\Rightarrow v \in [v]/\sim$ (Definition of $[v]/\sim$)

$\Rightarrow (\exists x) x \in [v]/\sim$ (EQ)

$\Rightarrow ([v]/\sim) \neq \emptyset$ (Lemma 3.5.8)

$\Rightarrow <[v]/\sim>$ exists (Definition of induced subgraph)

$\Rightarrow \omega(G) \geq 1$ ($<[v]/\sim>$ is a connected component in G) ■

Lemma 10.2.5: *Let $G_i = (V_i, E_i), 1 \leq i \leq \omega$, be the connected components of $G = (V, E)$. Then $\bigcup_{i=1}^{\omega} V_i = V$ and $\bigcup_{i=1}^{\omega} E_i = E$.*

Proof: Left as an exercise. ■

Lemma 10.2.6: *Let H and H' be two distinct connected components of a graph G=(V,E). Then,*

$$V_H \cap V_{H'} = \emptyset \text{ and } E_H \cap E_{H'} = \emptyset.$$

Proof: (Direct proof)

Let $H = <[u]/\sim>$ and $H' = <[v]/\sim>$ such that $H \neq H'$. $\cdots$(I)

Then $V_H = [u]/\sim$ and $V_{H'} = [v]/\sim$. $\cdots$(II)

Suppose $V_H \cap V_{H'} \neq \emptyset$. Then $([u]/\sim) \cap ([v]/\sim) \neq \emptyset$ ((II), Sub$_=$)

$\Rightarrow [u]/\sim = [v]/\sim$ ($[V_G]/\sim$ is a partition)

$\Rightarrow \langle [u]/\sim \rangle = \langle [v]/\sim \rangle$ (uniqueness of induced subgraph)

$\Rightarrow H = H'$ ((I), Sub$_=$), which contradicts $H \neq H'$.

Therefore, $V_H \cap V_{H'} = \emptyset$.

Suppose $E_H \cap E_{H'} \neq \emptyset$. Then, $\exists e \in E_H \cap E_{H'}$ (Lemma 3.5.8)

$\Rightarrow \{u, v\} \in E_H \cap E_{H'}$ (EI)

$\Rightarrow \{u, v\} \in E_H \wedge \{u, v\} \in E_{H'}$ (Definition of $\cap$)

$\Rightarrow u \in V_H \wedge u \in V_{H'}$ (Definition of E_H and $E_{H'}$)

$\Rightarrow u \in V_H \cap V_{H'}$ (Definition of $\cap$)

$\Rightarrow (\exists x) x \in V_H \cap V_{H'}$ (EQ)

$\Rightarrow V_H \cap V_{H'} \neq \emptyset$ (Lemma 3.5.8), which contradicts $V_H \cap V_{H'} = \emptyset$.

Therefore, $E_H \cap E_{H'} = \emptyset$. ■

Lemma 10.2.7: *Let H and H' be two distinct connected components of a graph G=(V,E). Then,*

$$(\forall u \in V_H)(\forall v \in V_{H'}) u \not\sim v.$$

Proof: (Proof by contradiction)

Suppose $(\exists u \in V_H)(\exists v \in V_{H'})u \sim v$.
Then $x \in V_H \wedge y \in V_{H'} \wedge x \sim y$ (EI)
$\Rightarrow x \in V_H$, $y \in V_{H'}$ and $x \sim y$. (E9,I2) . . . (I)
Let $V_H = [a]/\sim$ and $V_H = [b]/\sim$. $\cdots$ (II)
Then $x \in V_H \Rightarrow x \in [a]/\sim$ ($\text{Sub}_=$)
$\Rightarrow a \sim x$ (Definition of $[a]/\sim$) $\cdots$ (III)
$\Rightarrow a \sim x \wedge x \sim y$ ((I),(III),I6)
$\Rightarrow a \sim y$ ($\sim$ is transitive)
$\Rightarrow y \in [a]/\sim$ (Definition of $[a]/\sim$) $\cdots$ (IV)
But $y \in V_{H'} \Rightarrow y \in [b]/\sim$ ((II), $\text{Sub}_=$) $\cdots$ (V)
$\Rightarrow (y \in [a]/\sim) \wedge (y \in [b]/\sim)$ ((IV),(V),I6)
$\Rightarrow y \in ([a]/\sim) \cap ([b]/\sim)$ (definition of $\cap$)
$\Rightarrow (\exists w)(w \in ([a]/\sim) \cap ([b]/\sim))$ (EQ)
$\Rightarrow ([a]/\sim) \cap ([b]/\sim) \neq \emptyset$ (Lemma 3.5.8)
$\Rightarrow H \cap H' \neq \emptyset$ ((II), $\text{Sub}_=$) $\cdots$ (VI)
But by Lemma 10.2.6, $H \cap H' = \emptyset$ which contradicts (VI). ■

Theorem 10.2.8: *A graph G=(V,E) is disconnected iff* $\omega(G) \geq 2$.

Proof: $\Rightarrow$) G is disconnected
$\Rightarrow \exists u, v \in V$ such that $u \not\sim v$ (Definition of disconnected graph)
$\Rightarrow v \notin [u]/\sim$ (Definition of $[u]/\sim$) . . . (I)
But $v \sim v$ ($\sim$ is reflexive)
$\Rightarrow v \in [v]/\sim$ (Definition of $[v]/\sim$) . . . (II)
We thus have $v \in [v]/\sim \wedge v \notin [u]/\sim$ ((I),(II),I6)
$\Rightarrow (\exists x)(x \in [v]/\sim \wedge x \notin [u]/\sim)$ (EQ)
$\Rightarrow [v]/\sim \neq [u]/\sim$ (Lemma 3.2.2)
$\Rightarrow <[v]/\sim> \neq <[u]/\sim>$ (Definition of graph equality, =)
$\Rightarrow G$ has at least two connected components (Definition of connected components)
$\Rightarrow \omega(G) \geq 2$.

$\Leftarrow$) Let $\omega(G) \geq 2$. Then there exist two connected components H and H' of G such that $H \neq H'$.
Let $u \in V_H$ and $v \in V_{H'}$. . . . (I)
Then $H \neq H'$
$\Rightarrow (\forall x \in V_H)(\forall y \in V_{H'})x \not\sim y$. (Lemma 10.2.7)
$\Rightarrow u \in V_H \wedge v \in V_{H'} \Rightarrow u \not\sim v$. (UI)
$\Rightarrow u \not\sim v$ ((I),I6,I3) . . . (II)
Since $u \in V_H$, $v \in V_{H'}$ and $V_H, V_{H'} \subseteq V$,
We thus have $u \in V$ and $v \in V$ (Definition of $\subseteq$) . . . (III)
$\Rightarrow u \in V \wedge v \in V \wedge u \not\sim v$ ((II),(III),I6)
$\Rightarrow (\exists x \in V)(\exists y \in V)x \not\sim y$ (EQ)
$\Rightarrow G$ is disconnected. (Definition of disconnected graph) ■

Lemma 10.2.9: *Let $G_i = (V_i, E_i), 1 \leq i \leq \omega$, be the connected components of $G = (V, E)$ such that the order and size of G_i are p_i and q_i, respectively. Then,*

$$\sum_{i=1}^{\omega} p_i = |V| \text{ and } \sum_{i=1}^{\omega} q_i = |E|.$$

Proof: (By induction on ω)

(Induction basis) When $\omega = 1$, $G_1 \equiv G$.

It follows that $\sum_{i=1}^{\omega} p_i = \sum_{i=1}^{1} p_i = p_1 = |V|$ and $\sum_{i=1}^{\omega} q_i = \sum_{i=1}^{1} q_i = q_1 = |E|$.

(Induction hypothesis) Suppose the Lemma holds true for $\omega < m$.

(Induction step) Let $G = (V, E)$ be a graph with m connected components $G_i, 1 \leq i \leq m$, where $m \geq 2$.

Let $G' = (V', E')$ be the graph resulting from G after the connected component G_1 is removed from G. Therefore, G' consists of the connected components $G_i, 2 \leq i \leq m$, of G.

Since G' has $(m-2)+1 = m-1$ connected components, by the induction hypothesis,

$$\sum_{i=2}^{m} |V_i| = \sum_{i=2}^{m} p_i = |V'| \text{ and } \sum_{i=2}^{m} |E_i| = \sum_{i=2}^{m} q_i = |E'| \cdots\cdots \text{(I)}$$

By lemma 10.2.6, $V_1 \cap V_i = \emptyset$ and $E_1 \cap E_i = \emptyset$, for $2 \leq i \leq m$.

It follows that $\bigcup_{i=2}^{m} (V_1 \cap V_i) = \bigcup_{i=2}^{m} \emptyset$

$\Rightarrow \bigcup_{i=2}^{m} (V_1 \cap V_i) = \emptyset$ (Theorem 4.1.1(i))

$\Rightarrow V_1 \cap \bigcup_{i=2}^{m} V_i = \emptyset$ (Theorem 7.5.8(ii)) $\cdots\cdots$ (II)

But by Lemma 10.2.5, $V = \bigcup_{i=1}^{m} V_i = V_1 \cup \bigcup_{i=2}^{m} V_i$.

As a result, $|V| = \left| V_1 \cup \bigcup_{i=2}^{m} V_i \right|$

$= |V_1| + \left| \bigcup_{i=2}^{m} V_i \right|$ (Theorem 7.2.3), (II)

$= |V_1| + |V'|$ (Lemma 10.2.5)

$= p_1 + \sum_{i=2}^{m} p_i$ ((I), $\text{Sub}_=$)

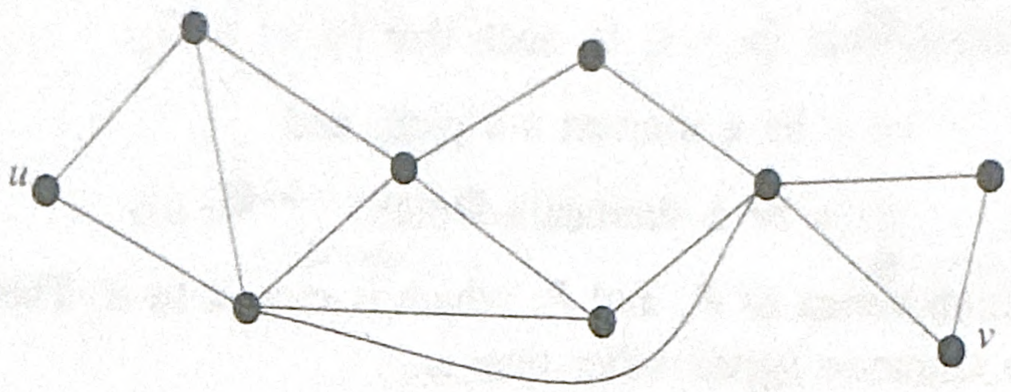
$= \sum_{i=1}^{m} p_i$

By a similar argument, we obtain $\sum_{i=1}^{m} q_i = |E|$. ■

10.3 Concepts related to Paths and Cycles

Definition: The ***length of a path*** (cycle, circuit, trail, walk, respectively) is the number of edges in the path (cycle, circuit, trail, walk, respectively). the ***distance*** between two vertices u and v in a graph G, ***$dist_G(u,v)$***, is the length of a shortest u-v path. The ***diameter*** of a graph, denoted by $diameter(G)$, is the longest distance in that graph, i.e. $diameter(G) = \max\{dist_G(u,v) | u, v \in V_G\})$. The ***girth*** of G is the length of a shortest cycle in G.

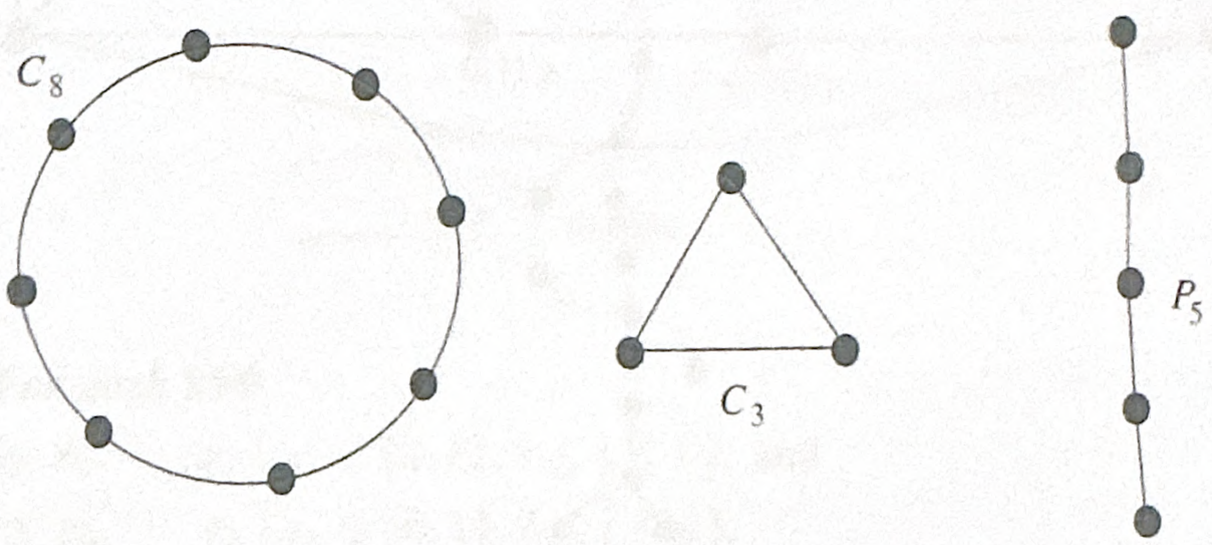
Example 10.3.8 In the following graph, $dist(u,v)=3$ which happens to be the diameter of the graph. The girth of the graph is 3.



Definition: A graph of order n which consists of exactly one cycle is called a ***cycle graph***. A cycle of order three is often called a ***triangle***. A graph of order n which consists of exactly one path is called a ***path graph***.

Since every two cycle (path, respectively) graphs of order n have identical graphical representation, they are isomorphic. We shall thus say *the* cycle (path, respectively) graph of order n and shall use C_n (P_n, respectively) to denote any such graph.

Figure 8 The cycle graphs C_3 and C_8, and the path graph C_5.



A characterization theorem of bipartite graphs

Theorem 10.3.10: *A graph is bipartite if and only if it contains no cycle of odd length.*

Proof: $\Rightarrow$) Let G be a bipartite graph with bipartition (X,Y).

Let $C : v_0v_1 \cdots v_{k-1}v_0$ be any cycle in G. The length of C is k.

Since $(v_i, v_{(i+1) \bmod k}) \in E_G, 0 \leq i < k$, $v_i \in X$ iff $v_{(i+1) \bmod k} \in Y$ . . . . (I)

In other words, the vertex following vertex v_i in cycle C belongs to Y iff v_i belongs to X

Without loss of generality, let $v_0 \in X$. Then by repeatedly applying (I), we obtain: "$v_i \in X$ iff i is even."

Since $v_{k-1} \in Y$, it follows that (k-1) is odd, or equivalently, k is even. i.e. C is of even length.

$\Leftarrow$) Without loss of generality, we assume G is connected.

Let s be any vertex in G.

Let $V_o = \{v \in V_G | dist_G(s,v)\ is\ odd\}$ and $V_e = \{v \in V_G | dist_G(s,v)\ is\ even\}$.

Since $\forall v \in V$, $dist_G(s,v)$ is unique, and is either even or odd but not both, we immediately obtain: $V_o \cup V_e = V_G$ and $V_o \cap V_e = \emptyset$.

We shall prove that there does not exist $u, v \in V_o$ or $u, v \in V_e$ such that $(u,v) \in E_G$.

Suppose to the contrary that $\exists u, v \in V_o$ such that $(u,v) \in E_G$.

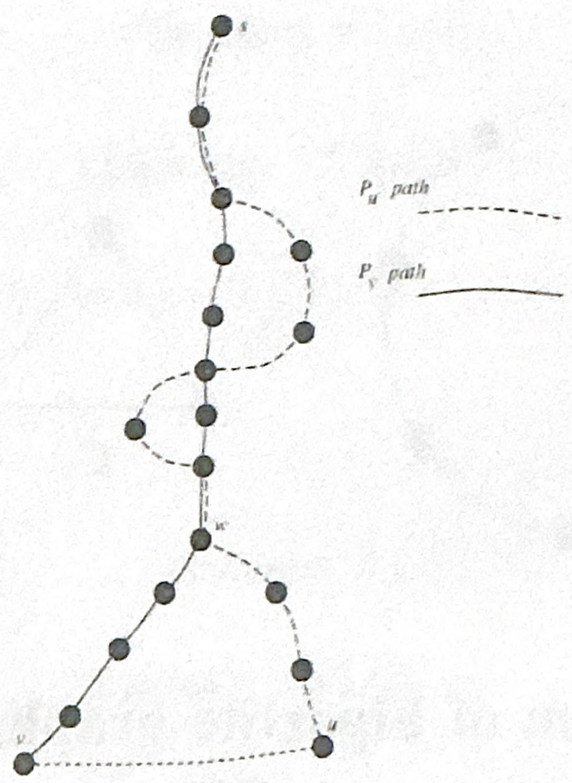
Let $P_u : s, v_1, v_2, \cdots, v_k, u$ be a shortest s-u path, and

$P_v : s, w_1, w_2, \cdots, w_l, v$ be a shortest s-v path.

Let ω be the common vertex of P_u and P_v which is closest to u. Then the ω-u path in P_u and the ω-v path in P_v has no common vertex other than ω.

Furthermore, neither the ω-u path nor the ω-v path could contain the edge (u,v), for otherwise, $|dist(s,u) - dist(s,v)| = 1$ which implies that $dist(s,u), dist(s,v)$ cannot be both odd. This contradicts the assumption that $u, v \in V_o$.

Hence, the ω-u path, the ω-v path and the edge (u,v) form a cycle in G whose length is $(dist(s,u) - dist(s,\omega)) + (dist(s,v) - dist(s,\omega)) + 1 = dist(s,u) + dist(s,v) - 2dist(s,\omega) + 1$ which is an odd number. i.e. G contains an odd cycle, a contradiction!



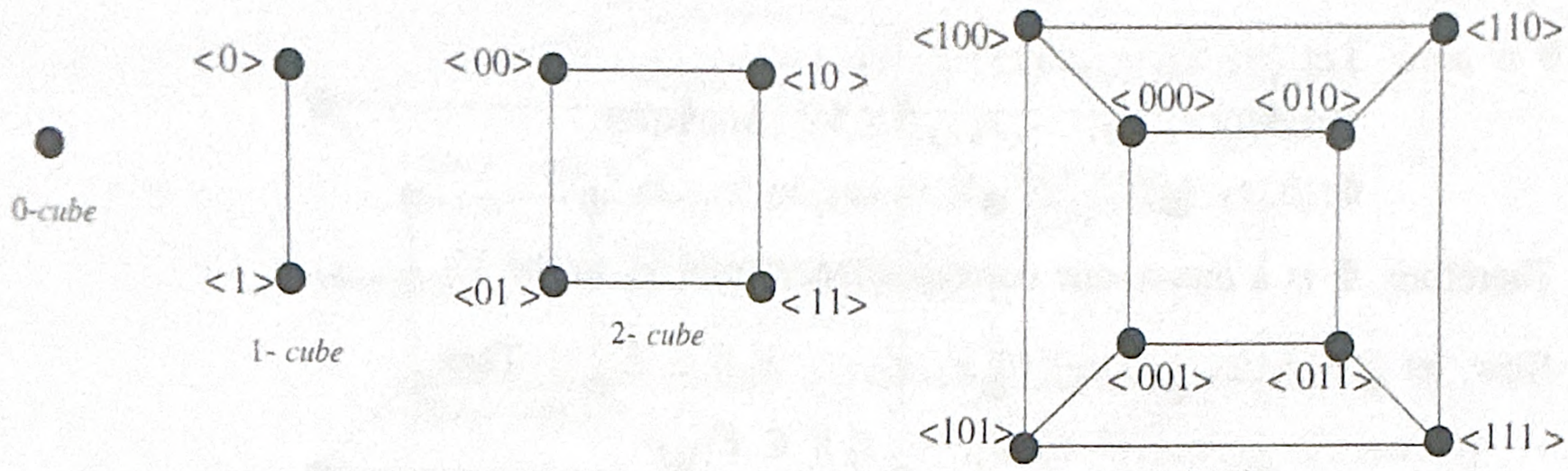
In a similar way, the assumption that there exists $u, v \in V_e$ such that $(u,v) \in E_G$ leads to a contradiction.

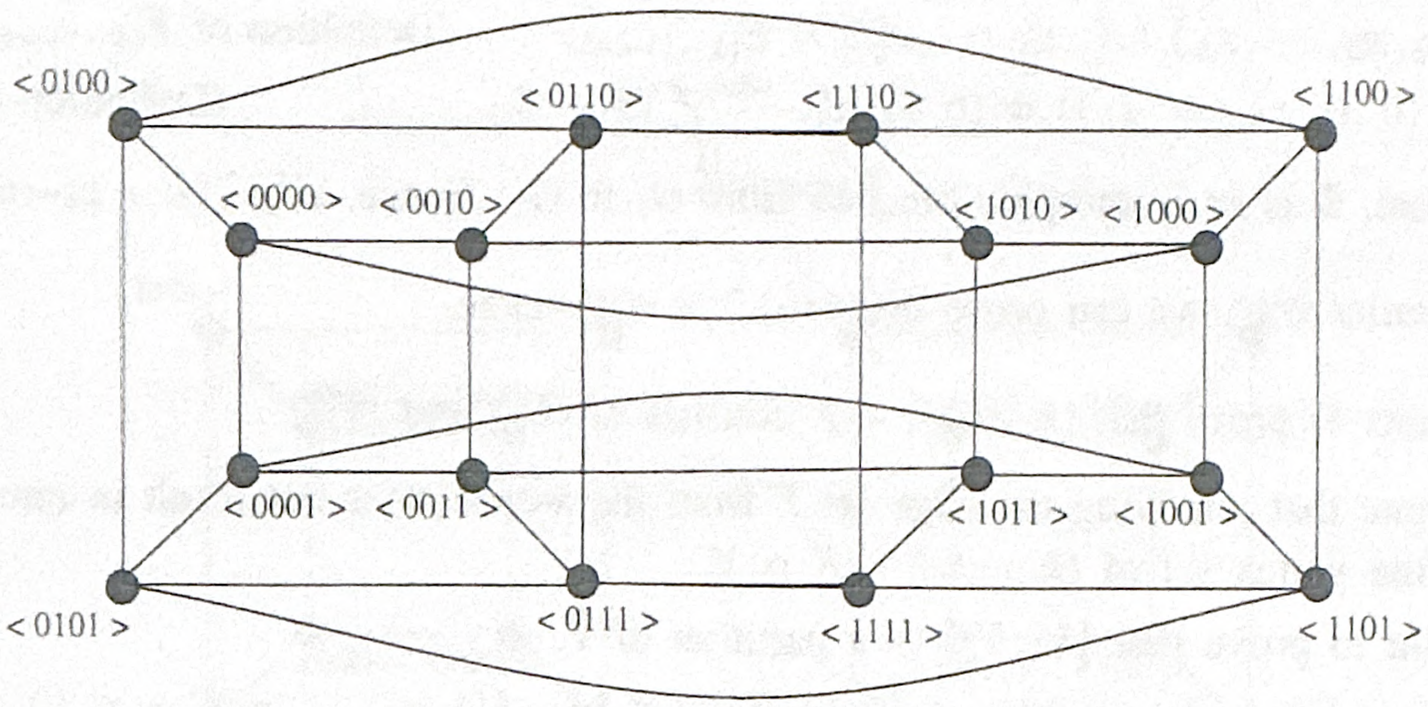
Hence, $\{V_0, V_e\}$ is a bipartition of G. i.e. G is a bipartite graph. ■

10.4 k-cube

Definition: The ***k-cube*** is the trivial graph when k=0 and is the graph whose vertex set is the set of k-tuples of 0's and 1's (i.e. $\{(x_1, x_2, \cdots, x_k) | x_i \in \{0,1\}, 1 \leq i \leq k\}$) in which two vertices are adjacent iff they differ in exactly one coordinate, when k>0.

Figure 9 The k-cubes, $0 \le k \le 4$.





Theorem 10.4.11: *For each k>0,*

let $V_1 = \{(0, x_2, x_3, \cdots, x_k) | x_i \in \{0,1\}, 2 \le i \le k\}$, and

$V_2 = \{(1, x_2, x_3, \cdots, x_k) | x_i \in \{0,1\}, 2 \le i \le k\}$.

Then $\langle V_1 \rangle \tilde{=} (k-1)$–*cube*; $\langle V_2 \rangle \tilde{=} (k-1)$–*cube*. Moreover,

let $F = \{\{(0, x_2, x_3, \ldots, x_k), (1, x_2, x_3, \ldots, x_k)\} \mid x_i \in \{0,1\}, 1 \le i \le k\}$, and

$(k$–*cube*$) - F$ denote the graph resulting from $(k$–*cube*$)$ after all the edges in F are removed.

Then $(k$–*cube*$) - F$ consists of $\langle V_1 \rangle$ and $\langle V_2 \rangle$.

Proof: Let V_{k-cube} and E_{k-cube} denote the vertex set and the edge set of the k-cube, and

$V_{\langle V_i \rangle}, E_{\langle V_i \rangle}$ denote the vertex set and edge set of $\langle V_i \rangle, 1 \le i \le 2$, respectively.

When $k = 1$, the theorem clearly holds true. Suppose $k \ge 2$.

Define $\Phi : V_1 \rightarrow V_{(k-1)-cube}$ such that

$$\Phi((0, x_2, x_3, \cdots, x_k)) = (x_2, x_3, \cdots, x_k).$$

We want to prove that Φ is an isomorphic function from $\langle V_1 \rangle$ to the (k-1)–cube.

(i) Φ is *one-to-one*:

$$\Phi((0, x_2, x_3, \cdots, x_k)) = \Phi((0, x_2', x_3', \cdots, x_k'))$$

$$\Rightarrow (x_2, x_3, \cdots, x_k) = (x_2', x_3', \cdots, x_k') \qquad (Definition\ of\ \Phi)$$

$\Rightarrow x_i = x_i', 2 \leq i \leq k$ (Lemma 7.5.9)

$\Rightarrow (0, x_2, x_3, \cdots, x_k) = (0, x_2', x_3', \cdots, x_k')$ (Lemma 7.5.9)

(ii) Φ is *onto*: Let $(x_1, x_2, \cdots, x_{k-1}) \in V_{(k-1)\text{–}cube}$.

Then $(0, x_1, x_2, \cdots, x_{k-1}) \in V_1$. Moreover,

$\Phi((0, x_1, x_2, \cdots, x_{k-1})) = (x_1, x_2, \cdots, x_{k-1})$.

Therefore, Φ is a one-to-one correspondence from V_1 to the $V_{(k-1)\text{–}cube}$.

Now, let $\{(0, x_2, x_3, \cdots, x_k), (0, x_2', x_3', \cdots, x_k')\} \in E_{\langle V_1 \rangle}$. Then,

$\{(0, x_2, x_3, \cdots, x_k), (0, x_2', x_3', \cdots, x_k')\} \in E_{\langle V_1 \rangle}$

$\Leftrightarrow \exists j, 2 \leq j \leq k, x_j = 1 - x_j' \text{ and } \forall i, 2 \leq i \leq k, i \neq j, x_i = x_i'$ (Definition of $E_{\langle V_1 \rangle}$)

$\Leftrightarrow \{(x_2, x_3, \cdots, x_k), (x_2', x_3', \cdots, x_k')\} \in E_{(k-1)\text{–}cube}$ (Definition of $E_{(k-1)\text{–}cube}$)

$\Leftrightarrow \{\Phi((0, x_2, x_3, \cdots, x_k)), \Phi((0, x_2', x_3', \cdots, x_k'))\} \in E_{(k-1)\text{–}cube}$ (Definition of Φ)

Therefore, Φ is an isomorphic function from G_1 to G_2. Hence, $\langle V_1 \rangle \tilde{=} (k-1)\text{–}cube$.

In a similar way, we can prove that $\langle V_2 \rangle \tilde{=} (k-1)\text{–}cube$.

It remains to prove that $(k\text{–}cube) - F$ consists of $\langle V_1 \rangle$ and $\langle V_2 \rangle$.

First, note that removing the edge set F from the k-cube does not result in removing any vertex. Therefore, the vertex set of $(k\text{–}cube) - F$ is V.

We want to prove that $\{V_1, V_2\}$ is a partition of V.

i. $V_1 \cup V_2 = \{(0, x_2, \cdots, x_k) \mid x_i \in \{0,1\}, 2 \leq i \leq k\} \cup \{(1, x_2, \cdots, x_k) \mid x_i \in \{0,1\}, 2 \leq i \leq k\}$

$= \{(x_1, x_2, \cdots, x_k) \mid x_i \in \{0,1\}, 1 \leq i \leq k\}$

$= V$

ii. Since $(0, 0, \cdots, 0) \in V_1$ and $(1, 0, \cdots, 0) \in V_2$, $V_1 \neq \emptyset$ and $V_2 \neq \emptyset$.

iii. Suppose $V_1 \cap V_2 \neq \emptyset$. Then, there exists $x \in (V_1 \cap V_2)$.

This implies that $x \in V_1 \wedge x \in V_2 \Rightarrow x = (0, x_2, \cdots, x_k) \wedge x = (1, x_2, \cdots, x_k)$ (Definitions of V_1, V_2)

$\Rightarrow (0, x_2, \cdots, x_k) = (1, x_2, \cdots, x_k)$ (= is transitive)

$\Rightarrow (0 = 1)$, a contradiction!

Therefore, $V_1 \cap V_2 = \emptyset$.

Hence, $\{V_1, V_2\}$ is a partition of V. This immediately implies that $\langle V_1 \rangle$ and $\langle V_2 \rangle$ have no common vertex and hence have no common edge.

Next, we observe that:

E_{k-cube}

$= \{\{(0, x_2, \cdots, x_i, \cdots, x_j, \cdots, x_k), (0, x_2, \cdots, x_i, \cdots, (1 - x_j), \cdots, x_k)\} | x_i, x_j \in \{0,1\}, 2 \leq i, j \leq k, i \neq j\}.$

$\cup \{\{(1, x_2, \cdots, x_i, \cdots, x_j, \cdots, x_k), (1, x_2, \cdots, x, \cdots, (1 - x_j), \cdots, x_k)\} | x_i, x_j \in \{0,1\}, 2 \leq i, j \leq k, i \neq j\}.$

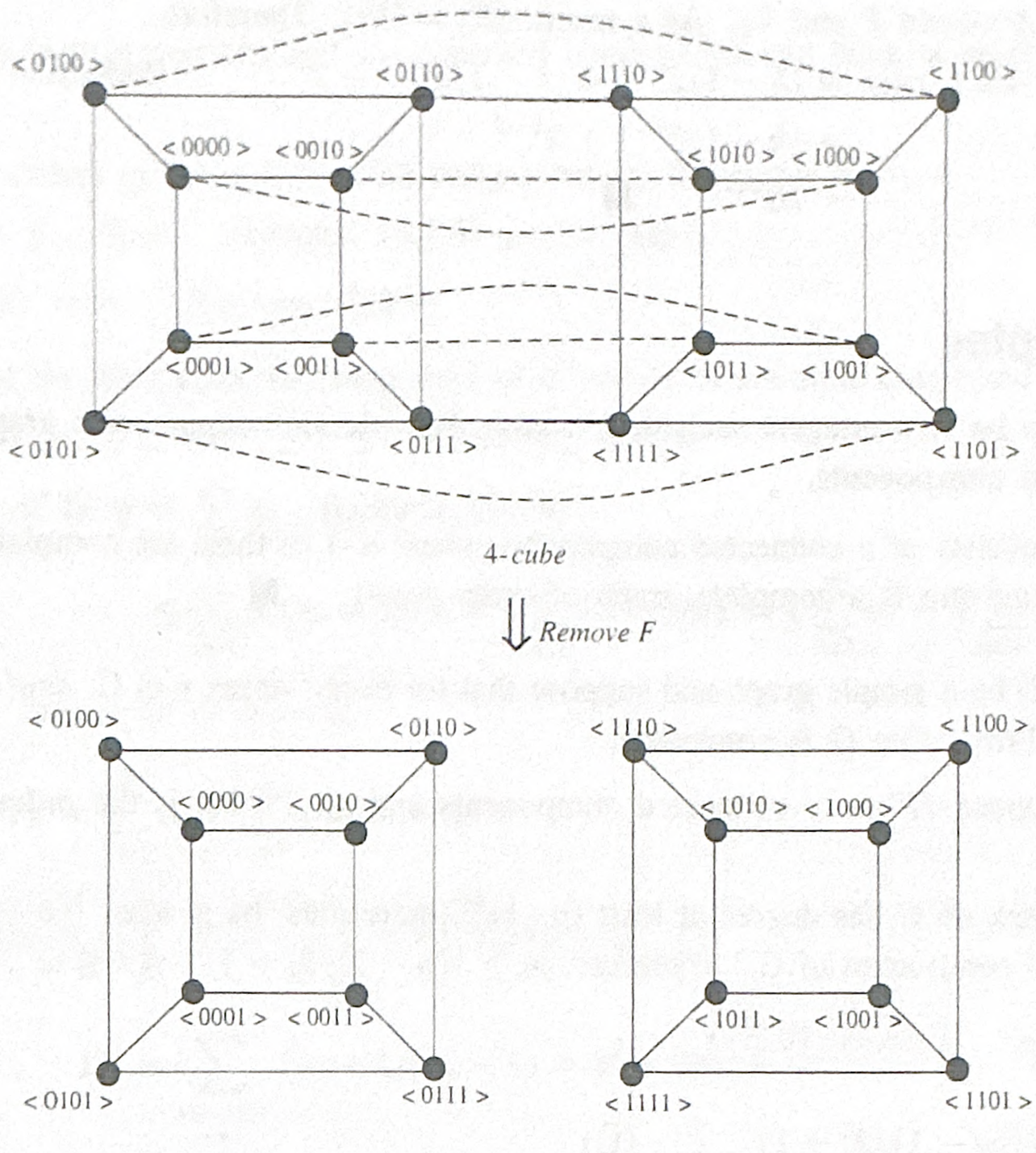
$\cup \{\{(0, x_2, x_3, \cdots, x_i \cdots, x_k), (1, x_2, x_3, \cdots, x_i, \cdots, x_k)\} | x_i \in \{0,1\}, 2 \leq i \leq k\}.$

$= E_{\langle V_1 \rangle} \cup E_{\langle V_2 \rangle} \cup F.$

Therefore, the edge set of $(k\text{–}cube) - F$ is $(E_{\langle V_1 \rangle} \cup E_{\langle V_2 \rangle} \cup F) - F = E_{\langle V_1 \rangle} \cup E_{\langle V_2 \rangle}$.

But, we have already shown that the vertex set of $(k\text{–}cube) - F$ is $V = V_1 \cup V_2$. So, by the definition of equality of graphs, the graph $(k\text{–}cube) - F$ consists of $\langle V_1 \rangle$ and $\langle V_2 \rangle$. ■

Figure 10.4.10 Decomposing the 4–*cube*. The dotted lines represent the edges in F.



The above theorem points out that we can decompose the k-cube into two (k-1)-cubes by removing the edge set F. Figure 10.4.10 illustrates the idea with the 4–cube. Therefore, the theorem provides us with a basis for using mathematical induction to prove properties of the *k-cubes*. The following is an example. Notice that the edge set F is the set of edges joining vertices in V_1 to vertices in V_2.

Example 10.4.9 Prove that the *k-cube* has order 2^k and size $k2^{k-1}$.

Solution: (By induction on k)

(*Induction basis*) The 0–*cube* is the trivial graph whose order is $1 = 2^0 = 2^k$ and whose size is

$$0 = 0 \times 2^{-1} = 0 \times 2^{0-1} = k2^{k-1}.$$

(*Induction hypothesis*) Suppose the (k-1)-*cube* has order 2^{k-1} and size $(k-1)2^{k-2}$.

(*Induction step*) Consider the *k-cube*($k \geq 1$). Let F be the set of edges defined in Theorem 10.4.11. Then the graph (k–*cube*)$-F$ consists of $\langle V_1 \rangle, \langle V_2 \rangle$ such that

$$\langle V_1 \rangle \tilde{=} (k-1) - cube \text{ and } \langle V_2 \rangle \tilde{=} (k-1) - cube.$$

By Theorem 9.1.2, and the induction hypothesis, $|V_1| = |V_{(k-1)-cube}| = 2^{k-1}$; and $|V_2| = |V_{(k-1)-cube}| = 2^{k-1}$.

The order of the *k-cube* is therefore $|V_1| + |V_2| = 2^{k-1} + 2^{k-1} = 2^k$.

The size of the *k-cube* = the size of $\langle V_1 \rangle$ + the size of $\langle V_2 \rangle$ + $|F|$

Since $F = \{\{(0, x_2, \cdots, x_k), (1, x_2, \cdots, x_k)\} \mid x_i \in \{0,1\}, 2 \leq i \leq k\}$, every edge in F has an end-vertex in V_1 and every vertex in V_1 is an end-vertex of F. This means that there is a one-to-one correspondence between F and V_1. As a result, $|F| = |V_1|$. Therefore,

$$\begin{aligned}\text{The size of the } k\text{-cube} &= (k-1)2^{k-2} + (k-1)2^{k-2} + 2^{k-1} \qquad \text{(Induction hypothesis)}\\ &= (k-1)2^{k-1} + 2^{k-1}\\ &= k2^{k-1} \quad \blacksquare\end{aligned}$$

10.5 Examples

1. Let n and p be two integers such that $1 \leq n \leq p$. Provide an example of a graph of order p and has n connected components.

Solution: G consists of n connected components where $n-1$ of them are complete graphs of order *one* and the remaining one is a complete graph of order p-n+1. ■

2. Let $G=(V,E)$ be a simple graph and suppose that for every vertex v in G, $deg(v) \geq (p-1)/2$ where $p = |V|$. Prove that G is connected.

Solution: Suppose G has ω connected components and p_i, $1 \leq i \leq \omega$, is the order of the ith connected component.

Since every vertex of G has degree at least $(p-1)/2$, there must be at least $\lceil (p-1)/2 \rceil + 1$ vertices in every connected component of G.[5] Therefore, $p_i \geq \lceil (p-1)/2 \rceil + 1, 1 \leq i \leq \omega$ . . . . (I)

$$\begin{aligned}\text{Now, } p &= \sum_{i=1}^{\omega} p_i \qquad \text{(Lemma 10.2.9)}\\ &\geq \sum_{i=1}^{\omega} (\lceil (p-1)/2 \rceil + 1) \qquad (\because \text{(I)})\\ &\geq \sum_{i=1}^{\omega} ((p-1)/2 + 1) \qquad (\lceil x \rceil \geq x)\\ &= \sum_{i=1}^{\omega} ((p+1)/2)\\ &= \omega(p+1)/2\end{aligned}$$

Therefore, $\omega \leq 2\frac{p}{p+1} < 2 \Rightarrow \omega \leq 1$

But $\omega \geq 1$, by Lemma 10.2.4.

Hence $\omega = 1$ ($\leq$ is antisymmetric), i.e. G is connected (Lemma 10.2.8). ■

3. Let G be a graph consisting of two connected components each of which is a complete graph and G is of even order p. Prove that the minimum size possible for G is $q = \frac{p^2-2p}{4}$. If G has this size, what does G look like?

Solution: Let the two connected components of G be of order x and $p-x$, respectively. Then,

$$\begin{aligned}q &= \frac{x(x-1)}{2} + \frac{(p-x)(p-x-1)}{2} \qquad \text{(Section 8.8, Example 6)}\\ &= \frac{(x^2-x+p^2-2px+x^2-p+x)}{2}\\ &= x^2 - px + \frac{(p^2-p)}{2} = \left(x - \frac{p}{2}\right)^2 - \frac{p^2}{4} + \frac{(p^2-p)}{2} = \left(x - \frac{p}{2}\right)^2 + \frac{(p^2-2p)}{4}\end{aligned}$$

q has the minimum value when $\left(x - \frac{p}{2}\right)^2 = 0$. Therefore, the minimum size possible for G is $\frac{p^2-2p}{4}$.

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[5] $\lceil x \rceil$ is the smallest integer greater than or equal to x; $\lfloor x \rfloor$ is the largest integer smaller than or equal to x, where x is a real number.

Since G has the minimum size when $\left(x - \frac{p}{2}\right)^2 = 0$, i.e. $x = \frac{p}{2}$. The orders of the two connected components of G are thus $\frac{p}{2}$ and $p - \frac{p}{2} = \frac{p}{2}$. i.e. G consists of two complete graphs of order $\frac{p}{2}$. ■

4. A simple graph with n vertices and ω connected components can have at most $(n-\omega)(n-\omega+1)/2$ edges.

Solution: Let the orders of the ω connected components of the graph be $n_1, n_2, \cdots, n_\omega$.

Then $n_i \geq 1, 1 \leq i \leq \omega$. (Lemma 10.2.4) . . . . (I)

Moreover, $\sum_{i=1}^{\omega} n_i = n$. (Lemma 10.2.9)

Let q_i be the size of the ith connected component of G (which is a simple connected graph).

Then $q_i \leq n_i(n_i - 1)/2$, for $1 \leq i \leq \omega$ (Corollary 8.4.3).

Therefore, the size of G, $q = \sum_{i=1}^{\omega} q_i$ (Lemma 10.2.9)

$$\leq \sum_{i=1}^{\omega} n_i(n_i - 1)/2 = \frac{1}{2}\sum_{i=1}^{\omega}\left(n_i^2 - n_i\right) = \frac{1}{2}\sum_{i=1}^{\omega} n_i^2 - \frac{1}{2}\sum_{i=1}^{\omega} n_i = \frac{1}{2}\sum_{i=1}^{\omega} n_i^2 - \frac{n}{2}$$

But $\sum_{i=1}^{\omega} n_i^2 \leq n^2 - 2n\omega + \omega^2 + 2n - \omega$.

[**Proof:** $\sum_{i=1}^{\omega}(n_i - 1) = \sum_{i=1}^{\omega} n_i - \sum_{i=1}^{\omega} 1 = n - \omega$.

$$\therefore \left(\sum_{i=1}^{\omega}(n_i - 1)\right)^2 = (n-\omega)^2 = n^2 - 2n\omega + \omega^2$$

$$\Rightarrow \sum_{i=1}(n_i - 1)^2 + \sum_{\substack{1 \leq i,j \leq \omega \\ i \neq j}}(n_i - 1)(n_j - 1) = n^2 - 2n\omega + w^2$$

$$\Rightarrow \sum_{i=1}^{\omega}(n_i - 1)^2 \leq n^2 - 2n\omega + \omega^2 \qquad \left(\because \text{(I)} \Rightarrow \sum_{\substack{1 \leq i,j \leq \omega \\ i \neq j}}(n_i - 1)(n_j - 1) \geq 0\right)$$

$$\Rightarrow \sum_{i=1}^{\omega}\left(n_i^2 - 2n_i + 1\right) \leq n^2 - 2n\omega + \omega^2$$

$$\Rightarrow \sum_{i=1}^{\omega} n_i^2 - 2\sum_{i=1}^{\omega} n_i + \sum_{i=1}^{\omega} 1 \leq n^2 - 2n\omega + \omega^2$$

$$\Rightarrow \sum_{i=1}^{\omega} n_i^2 - 2n + \omega \leq n^2 - 2n\omega + \omega^2$$

Therefore, $\sum_{i=1}^{\omega} n_i^2 \leq n^2 - 2n\omega + \omega^2 + 2n - \omega$]

Hence, $q \leq \frac{1}{2}\sum_{i=1}^{\omega} n_i^2 - \frac{n}{2} \leq \frac{1}{2}\left(n^2 - 2n\omega + \omega^2 + 2n - \omega\right) - \frac{n}{2} = \frac{1}{2}(n-\omega)(n-\omega+1)$. ■

5. Prove that a simple graph G with n vertices must be connected if it has more than $(n-1)(n-2)/2$ edges.

Solution: (*Proof by Contradiction:*) suppose G is disconnected. Let G has size q and ω connected components.

Then $\omega \geq 2$ (Lemma 10.2.8)

$\Rightarrow -\omega \leq -2 \Rightarrow 0 \leq n - \omega \leq n - 2$ (because $n \geq \omega$) . . . . . (I)

$\Rightarrow 0 < n - \omega + 1 \leq n - 2 + 1 = n - 1$ ..... (II)

From (I) and (II), we have $(n-\omega)(n-\omega+1)/2 \leq (n-1)(n-2)/2$

From Example 4, we have the size of G, $q \leq (n-\omega)(n-\omega+1)/2$

On the other hand, it is given that $(n-1)(n-2)/2 < q$.

We therefore have, $q<q$, a contradiction. Hence, G is connected. ■

6. Show that :

(a) the size of $K_{m,n}$ is mn.

(b) if G is a simple bipartite graph of size q, Then $q \leq \frac{p^2}{4}$.

Solution: *(a)* Let (X,Y) be the bipartition of $K_{m,n}$ with $|X| = m; |Y| = n$. By Theorem 8.4.1,

$$\sum_{v \text{ in } K_{m,n}} deg(v) = 2q$$

$$\Rightarrow \sum_{v \in X} deg(v) + \sum_{v \in Y} deg(v) = 2q \quad ((X,Y) \text{ is the bipartition of } K_{m,n})$$

$$\Rightarrow \sum_{v \in X} n + \sum_{v \in Y} m = 2q \quad (deg(v) = m, \forall v \in Y;\ deg(v) = n, \forall v \in X)$$

$$\Rightarrow mn + nm = 2q \quad (|X| = m; |Y| = n)$$

$$\Rightarrow 2mn = 2q$$

$$\Rightarrow q = mn$$

(b) Let (X,Y) be a bipartition of G with $|X| = m$. Then $|Y| = p - m$.

Since the size of G is not greater than that of $K_{m,p-m}$, we have,

$$q \leq m(p-m) \qquad (\text{Part}(a))$$

$$= pm - m^2 = -\left(m^2 - 2\left(\tfrac{p}{2}\right)m + \left(\tfrac{p}{2}\right)^2 - \left(\tfrac{p}{2}\right)^2\right)$$

$$= -\left(m - \tfrac{p}{2}\right)^2 + \left(\tfrac{p}{2}\right)^2 \leq 0 + \left(\tfrac{p}{2}\right)^2 = \tfrac{p^2}{4}$$ ■

7. Let $G_1 \tilde{=} G_2$. Prove that G_1 is connected iff G_2 is connected.

Solution: $\Rightarrow$) Suppose G_1 is connected.

Since $G_1 \tilde{=} G_2$, $\exists \Phi : V_{G_1} \underset{onto}{\overset{1-1}{\rightarrow}} V_{G_2}$ such that $\{a,b\} \in E_{G_1} \Leftrightarrow \{\Phi(a), \Phi(b)\} \in E_{G_2}$.

Let $x, y \in V_{G_2}$. Since, Φ is onto, $\exists u, v \in V_{G_1}$ such that $\Phi(u) = x$ and $\Phi(v) = y$.

As G_1 is connected, there exists an u-v path, $(u =)w_1\ w_2 \cdots w_k (= v)$, in G_1.

Since $\{w_i, w_{i+1}\} \in E_{G_1}, 1 \leq i < k$, by the definition of Φ, $\{\Phi(w_i), \Phi(w_{i+1})\} \in E_{G_2}, 1 \leq i < k$.

This implies that $(x =)\Phi(w_1)\ \Phi(w_2) \cdots \Phi(w_k)(= y)$ is an x-y path in G_2.

Therefore, G_2 is connected.

$\Leftarrow$) Similar to the $\Rightarrow$) part. ■

8. Prove that the *k-cube* is connected.

Solution: (By induction on k)

(*Induction basis*) When k=0. The 0–*cube* is the trivial graph which is connected.

(*Induction hypothesis*) Suppose the *(k*-1*)-cube*($k \geq 1$) is connected.

(*Induction step*) Consider the *k-cube*.

Let F be the edge set defined in Theorem 10.4.11, then the graph $(k\text{–}cube) - F$ consists of $\langle V_0 \rangle$ and $\langle V_1 \rangle$ such that $\langle V_0 \rangle \simeq (k-1) - cube$ and $\langle V_1 \rangle \simeq (k-1) - cube$.

By the induction hypothesis, the *(k-1)-cube* is connected. By Example 7, $\langle V_0 \rangle$ and $\langle V_1 \rangle$ are also connected.

Now, let u, v be any two vertices in the *k-cube*.

(i) $u, v \in V_0$ or $u, v \in V_1$: then u, v are connected in $\langle V_0 \rangle$ or $\langle V_1 \rangle$, and hence in the *k-cube*.

(ii) $u \in V_0$ and $v \in V_1$: Let $\{u, \overline{u}\}$ be the edge in F of which u is an end-vertex. Then, $\overline{u} \in V_1$. Since $\langle V_1 \rangle$ is connected, there exists a $\overline{u} - v$ path in $\langle V_1 \rangle$. But then the $\{u, \overline{u}\}$ edge and the $\overline{u} - v$ path form an u-v path in the *k-cube*. Therefore, vertices u and v are connected in the *k-cube*.

Hence, the *k-cube* is connected. ■

9. Prove that in any simple connected graph $G=(V,E)$, if G is not complete, then G has three vertices u, v, w such that $\{u, v\} \in E$, $\{v, w\} \in E$, and $\{u, w\} \notin E$.

Solution: (*Proof by contradiction*) Suppose G is a simple connected graph which is not a complete graph and in which there does not exist vertices u, v, w such that $\{u, v\} \in E$, $\{v, w\} \in E$, and $\{u, w\} \notin E$.

Then $\sim (\exists u, v, w \in V)(\{u, v\} \in E \wedge \{v, w\} \in E \wedge \{u, w\} \notin E)$

$\Rightarrow (\forall u, v, w \in V) \sim (\{u, v\} \in E \wedge \{v, w\} \in E \wedge \{u, w\} \notin E)$ (FE8$_\in$)

$\Rightarrow (\forall u, v, w \in V)(\sim (\{u, v\} \in E \wedge \{v, w\} \in E) \vee \sim \{u, w\} \notin E)$ (E16)

$\Rightarrow (\forall u, v, w \in V)(\sim (\{u, v\} \in E \wedge \{v, w\} \in E) \vee \{u, w\} \in E)$ (E15)

$\Rightarrow (\forall u, v, w \in V)(\{u, v\} \in E \wedge \{v, w\} \in E \Rightarrow \{u, w\} \in E)$ (E18) . . . . . (I)

G is not complete implies that there exists a pair of vertices u and v in G such that $\{u, v\} \notin E$.

G is connected implies that there exists an u-v path in G. Let P:$(u =)w_0, w_1, w_2, \cdots, w_k (= v)$ be a shortest u-v path.

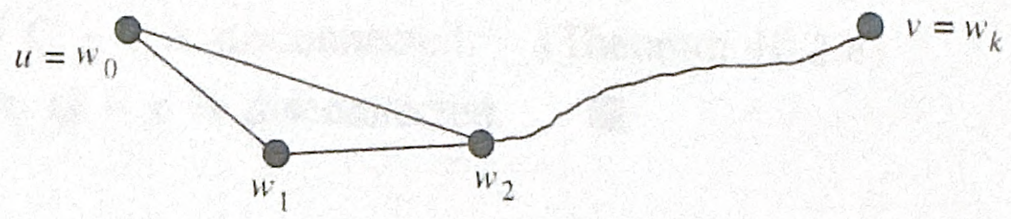
Since $\{u, v\} \notin E$, the u-v path P must contain at less two edges which implies that $k \geq 2$.

We thus have $\{u, w_1\} \in E$ and $\{w_1, w_2\} \in E$

$\Rightarrow \{u, w_1\} \in E \wedge \{w_1, w_2\} \in E$ (I6)

$\Rightarrow \{u, w_1\} \in E \wedge \{w_1, w_2\} \in E$ and $\{u, w_1\} \in E \wedge \{w_1, w_2\} \in E \Rightarrow \{u, w_2\} \in E$ ((I),UI)

$\Rightarrow (u, w_2) \in E$ (I3)



But than $(u =)w_0, w_2, w_3, \cdots, w_k (= v)$ is a u-v path of length $k-1$ which is shorter than the shortest u-v path P, a contradiction! ■.

Connect 10.6 Exercises

1. Prove that the relation $\sim$ is an equivalence relation.
2. Prove that if $G = (V, E)$ is disconnected, then $\overline{G} = (\overline{V}, \overline{E})$ is connected.
3. Prove that the distance between two vertices has the following properties:
   (i) $dist(u, v) = 0$ iff $u = v$

(ii) $dist(u, v) = dist(v, u)$

(iii) $dist(u, w) \leq dist(u, v) + dist(v, w)$

4. Prove that if G is a simple graph and $\delta(G) \geq k$, then G has a path of length k.
5. Let G and H be two simple graphs such that $G \cong H$. Prove that $\forall l \geq 0$, the number of cycles of length l in G is equal to the number of cycles of length l in H.
6. Prove that if $G = (V, E)$ has diameter greater than 3, then $\overline{G} = (\overline{V}, \overline{E})$ has diameter less than 3.
7. Let $G = (V, E)$ be such that $G \cong \overline{G}$. Prove that:

   (a) G is connected.

   (b) $diameter(G) \leq 3$

8. Prove that every two longest paths in a connected graph have at least one common vertex.
9. Prove that every k-cube is a bipartite graph.
10. Let $G = (V, E)$ be a connected simple graph such that G contains no C_3 or P_4 as an induced subgraph. Prove that G is a complete bipartite graph.